

LAND REGISTRATION DATABASE SYSTEM

Final Project Report

Database Platform	PostgreSQL
Database Interface	psql
Project Type	Relational Database Design and Implementation
Final Stage	Day 21 — Presentation and Submission
Prepared By	_____
Course / Programme	_____
Institution	_____
Submission Date	_____

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1. Introduction

The Land Registration Database System is a PostgreSQL-based relational database project designed to organize and manage information related to land registration. The system brings together data about landowners, land parcels, geographical areas, registration applications, payments, titles, ownership transfers and disputes.

The project demonstrates practical database concepts including relational modelling, entity relationships, primary keys, foreign keys, constraints, SQL queries, role-based security, reporting and end-to-end system testing.

2. Problem Statement

Land registration requires several records that are related to one another. A landowner may submit an application for a parcel, make a payment, receive a title and later transfer ownership. A parcel may also be associated with a dispute. When these records are not properly connected, searching, reporting and maintaining accurate information becomes difficult.

The problem addressed by this project is the need for a structured relational database that stores these records consistently, maintains valid relationships between them and supports secure access and administrative reporting.

3. Project Objectives

- Design and implement a Land Registration Database using PostgreSQL.
- Register and manage landowners.
- Register and manage land parcels and their locations.
- Manage land registration applications.
- Record payments associated with applications.
- Create and manage land titles.
- Track ownership transfers.
- Record and monitor land disputes.
- Enforce data integrity through database constraints and relationships.
- Implement role-based access control.
- Produce administrative and operational reports.

4. System Scope

The project focuses on the database layer of a land registration system. It does not depend on a web or mobile user interface; instead, the database is managed and demonstrated through PostgreSQL and the `psql` command-line interface.

Included	Not the main focus of this project
Landowner registration	Public-facing web interface
Parcel and location records	Mobile application
Applications and payments	Online payment gateway integration
Title creation	GIS or map visualization
Transfers and disputes	Document scanning workflow
Security roles and permissions	Full production deployment infrastructure
SQL reports and live testing	Automated external API integration

5. Methodology and Tools

The project followed a practical database development approach. Requirements were identified, entities and relationships were modelled, the schema was implemented, sample data and SQL operations were tested, roles were created, reports were generated and a complete end-to-end workflow was demonstrated.

Tool / Technology	Purpose
PostgreSQL	Relational database management system
psql	Command-line interface used to execute and demonstrate SQL
SQL	Database definition, manipulation, querying and security
ERD	Visual representation of entities and relationships

6. Entity Relationship Diagram

The ERD below presents the main entities used by the Land Registration Database and the relationships between them.

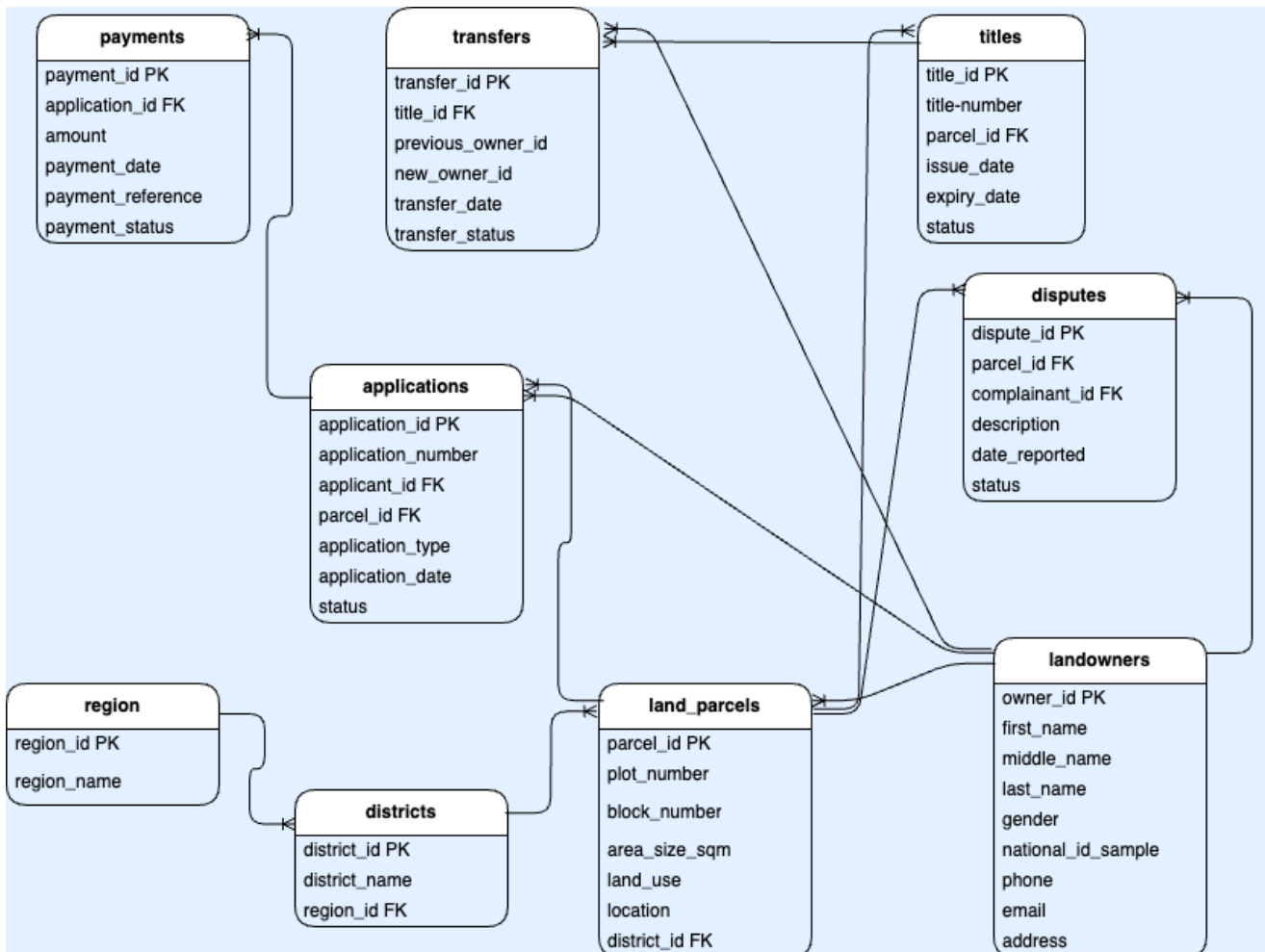


Figure 1: Entity Relationship Diagram for the Land Registration Database.

7. Database Schema

The implemented database is organized around the following core entities. Each table represents a specific part of the land registration process.

Table	Purpose
landowners	Stores personal and contact details for landowners.
regions	Stores regional geographical information.
districts	Stores districts and links each district to a region.
land_parcel	Stores plot, block, area, land use, location and ownership information.
applications	Stores land-related applications and their processing status.
payments	Stores payments associated with applications.
titles	Stores title information associated with land parcels.
transfers	Stores the history of ownership transfers.
disputes	Stores disputes associated with parcels and complainants.

8. Key Relationships and Data Integrity

The relational structure connects records through primary and foreign keys. Important relationships include:

- A region can contain multiple districts.
- A district can contain multiple land parcels.
- A landowner can be associated with land parcels and applications.
- An application is linked to an applicant and a parcel.
- A payment is linked to an application.
- A title is linked to a land parcel.
- A transfer is linked to a title and identifies previous and new owners.
- A dispute is linked to a parcel and a complainant.

Constraints Used

- Primary keys provide unique identifiers for records.
- Foreign keys prevent invalid relationships between tables.
- UNIQUE constraints prevent duplication of selected identifying values.
- CHECK constraints can restrict invalid statuses, amounts and other values.

9. SQL Queries and Operations

SQL queries were used to retrieve, update and verify information throughout the project. Examples include searching landowners and joining related tables for reports.

```
-- Search by first name
SELECT *
FROM landowners
WHERE first_name ILIKE '%John%';

-- Search by National ID
SELECT *
FROM landowners
WHERE national_id_sample = 'TZ-100001';

-- Search by phone number
SELECT *
FROM landowners
WHERE phone = '+255712000001';
```

Example of a relationship query:

```
SELECT
    a.application_number,
    lo.first_name || ' ' || lo.last_name AS applicant_name,
    lp.plot_number,
    a.application_type,
    a.application_date,
    a.status
FROM applications a
JOIN landowners lo
    ON a.applicant_id = lo.owner_id
JOIN land_parcel lp
    ON a.parcel_id = lp.parcel_id;
```

10. Security and Access Control

The project uses role-based access control to demonstrate database security. Different users can be given different permissions according to their responsibilities.

Role	Purpose
land_admin	Administrative access for managing database objects and records.
land_officer	Operational access for permitted registration and processing activities.
land_viewer	Read-only access for viewing data and reports.

```
GRANT SELECT ON ALL TABLES IN SCHEMA public TO land_viewer;

REVOKE INSERT, UPDATE, DELETE
ON ALL TABLES IN SCHEMA public
FROM land_viewer;
```

11. Final Reports

Day 19 focused on producing useful SQL reports from the database. The following reports were included:

Report	Purpose
Landowners Report	Lists registered landowners and their details.
Parcels by District	Shows land parcels organized by district.
Active Titles	Shows titles with ACTIVE status.
Pending Applications	Shows applications awaiting processing.
Payments	Shows payments linked to applications and applicants.
Transfer History	Shows previous and new owners for recorded transfers.

The report queries were saved together in the file **day19_final_reports.sql** so that they can be executed as part of the final demonstration.

12. Full System Testing

Day 20 tested one complete fictional workflow from owner registration to title creation. Additional tests were performed for a transfer, a dispute and database constraints.

Test Step	Expected Result
Landowner registration	A new owner record is created successfully.
Parcel registration	A parcel is linked to the correct district and owner.
Application creation	The application links the applicant and parcel.
Payment	A payment is linked to the correct application.
Approval	The application status is updated.
Title creation	An ACTIVE title is linked to the parcel.
Transfer	Previous and new owners are recorded.
Dispute	A dispute is linked to a parcel and complainant.
UNIQUE test	Duplicate identifying data is rejected.
Foreign-key test	Invalid related IDs are rejected.

13. Live Demonstration in psql

The database can be demonstrated live through the PostgreSQL command-line interface. The exact username and database name should be replaced with the values used in the local installation.

```
# Connect to the database
psql -U YOUR_USERNAME -d land_registration_db

-- Show tables
\dt

-- Describe a table
\d landowners

-- View data
SELECT * FROM landowners;

-- Run Day 19 reports
\i day19_final_reports.sql

-- Run Day 20 full test
\i day20_full_test.sql
```

During the final presentation, the recommended demonstration order is: show the database tables, inspect selected table structures, run example queries, demonstrate roles or permissions, execute the Day 19 reports and finally execute the Day 20 full test.

14. Results and Achievements

- A relational database structure was designed for the land registration domain.
- Core entities and relationships were represented using an ERD.
- Tables were implemented for owners, parcels, locations, applications, payments, titles, transfers and disputes.
- SQL queries were used for searching, joining and reporting.
- Role-based permissions were included to demonstrate security.
- Six final reports were prepared.
- A complete end-to-end fictional registration workflow was tested.
- Transfer and dispute scenarios were tested.
- Constraint and relationship verification was included in the full test.

15. Conclusion

The Land Registration Database System demonstrates the use of PostgreSQL to model and manage a connected land registration workflow. The project integrates information about landowners, parcels, applications, payments, titles, transfers and disputes through relational design.

The use of primary keys, foreign keys, constraints and role-based access control supports data integrity and security. SQL reports provide useful views of the stored data, while the full system test demonstrates the workflow from initial registration through payment, approval, title creation, transfer and dispute handling.

16. Project Deliverables and Folder Structure

The complete project submission should contain the database, SQL scripts, ERD, presentation and this final report.

```
Land_Registration_Project/
■
■■■ README.txt
■
■■■ database/
■   ■■■ land_registration_db.sql
■
■■■ sql/
■   ■■■ day19_final_reports.sql
■   ■■■ day20_full_test.sql
■
■■■ diagrams/
■   ■■■ land_registration_erd.png
■
■■■ presentation/
■   ■■■ Land_Registration_Day21_Presentation.pptx
■
■■■ report/
    ■■■ Land_Registration_Final_Project_Report.pdf
```

Final Submission Checklist

- Complete PostgreSQL database dump or SQL schema/data file
- ERD diagram
- Day 19 final reports SQL file
- Day 20 full system test SQL file
- Day 21 PowerPoint presentation
- Final project report PDF